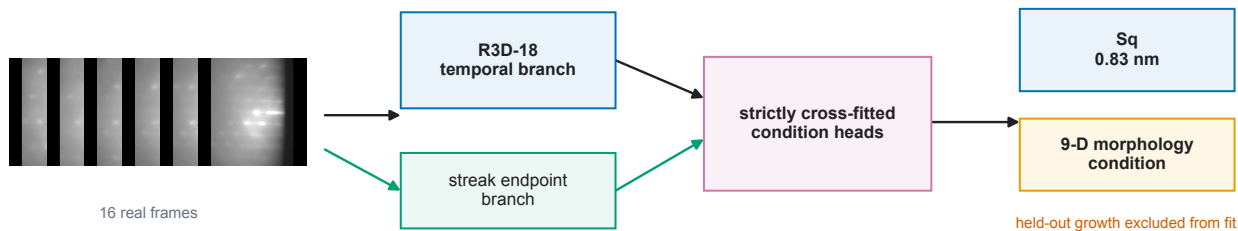


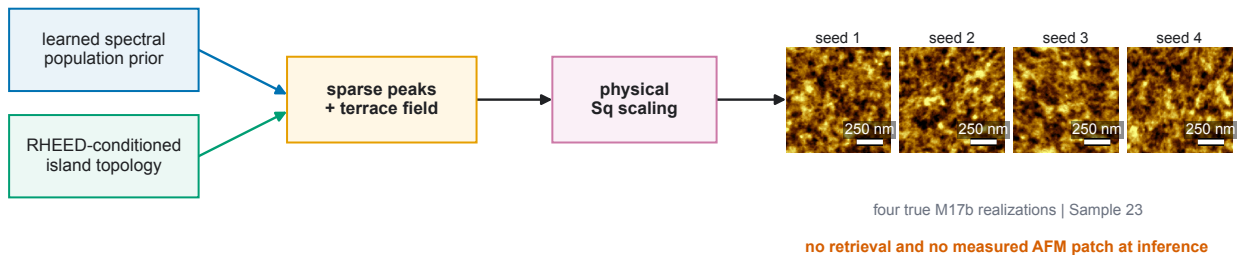
a

## RHEED representation and condition prediction



b

## M17b topology-conditioned sparse-peak terrace generator



c

## Growth-level leave-one-out validation



27 outer folds | growth groups are the leakage boundary | operator-invalid growth 6081 excluded before fitting